

Bio-Inspired Edge Detection: A Semester-Long Investigation of Biological Principles, Architectural Constraints, and Comparative Evaluation

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Abstract

Edge detection remains one of the most fundamental problems in computer vision, serving as the foundation for tasks such as segmentation, object recognition, scene understanding, and autonomous perception. While modern deep learning-based edge detectors achieve strong quantitative performance, they are often data-hungry, computationally expensive, and difficult to interpret.

This semester-long project investigates bio-inspired edge detection models, which draw inspiration from the human visual system to introduce explicit structural and functional inductive biases. The work initially focused on training a fully bio-inspired architecture, XYW-Net, but practical constraints related to dataset scale, training time, and the absence of pretrained models led to a methodological pivot.

The final experimental framework evaluates bio-inspired approaches—XDoG and the pretrained DEXINED model—against established deep learning baselines such as HED and RCF. This report documents the theoretical motivation, architectural analysis, experimental decisions, and scientific reasoning behind this transition, and argues that bio-inspired vision represents a promising and necessary direction for future edge detection research.

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1 Introduction

Edge detection is a core operation in computer vision, aiming to identify significant intensity discontinuities that correspond to object boundaries, surface changes, or structural transitions within an image. Reliable edge information supports higher-level tasks such as image segmentation, object recognition, depth estimation, and visual tracking.

Historically, edge detection was addressed using handcrafted operators such as Sobel, Prewitt, Laplacian of Gaussian, and Canny. These methods rely on gradient computations and fixed thresholds, making them sensitive to noise, illumination changes, and parameter tuning.

The emergence of deep learning introduced data-driven edge detection models that learn hierarchical feature representations from large annotated datasets. While these approaches significantly improved performance, they also introduced new challenges related to data requirements, computational cost, and lack of interpretability.

In parallel, bio-inspired vision models seek to replicate principles observed in biological visual systems, particularly the human retina and early visual cortex. This project explores whether such principles can provide meaningful advantages in edge detection under realistic constraints.

2 Edge Detection as a Vision Primitive

Edges are not merely low-level features; they encode essential structural information about the visual world. Psychophysical and neurophysiological studies show that biological vision systems prioritize edge and contrast information at very early stages of processing.

From a computational perspective, edges offer:

- A compact representation of scene geometry
- Reduced sensitivity to texture and illumination
- A bridge between pixel-level data and semantic understanding

This explains why edge detection remains relevant even in the era of end-to-end deep learning.

3 Limitations of Purely Data-Driven Models

Despite their success, modern deep learning-based edge detectors exhibit several limitations:

- Dependence on large-scale annotated datasets

- High computational and energy cost
- Sensitivity to domain shifts
- Limited interpretability of learned features

These limitations motivate the exploration of architectures that embed prior knowledge directly into their structure rather than learning everything from data.

4 Biological Vision as Inspiration

The human visual system performs robust visual processing under strict constraints on energy, bandwidth, and latency. A significant portion of this processing occurs in the retina, where signals are transformed through multiple specialized cell types before reaching the visual cortex.

Key biological mechanisms relevant to edge detection include:

- Center-surround receptive fields
- ON/OFF contrast pathways
- Orientation and direction selectivity
- Multi-scale hierarchical processing

Bio-inspired edge detectors attempt to translate these principles into computational models.

5 Bio-Inspired Edge Detection Paradigms

Bio-inspired edge detection methods can be broadly categorized into:

- Non-learning, filter-based models
- Hybrid models combining biological principles with learning
- Fully bio-inspired deep architectures

This project explores representatives of the first two categories in depth.

6 Initial Focus: XYW-Net

XYW-Net is a biologically motivated deep architecture designed to model parallel retinal processing pathways. Its encoder consists of three streams that process horizontal, vertical, and directional information, inspired by biological visual channels.

The model aims to produce semantically meaningful edges through long-term training and multi-scale refinement.

7 Practical Limitations of XYW-Net

Despite its theoretical appeal, training XYW-Net from scratch proved infeasible within the scope of this semester. The main limiting factors were:

- Extremely long training times
- Requirement for large and diverse datasets
- Absence of pretrained models
- Sensitivity to preprocessing and decoder design

Continuing training under these constraints would not produce reliable or scientifically valid results.

8 Methodological Pivot

Rather than forcing incomplete experiments, the project adopted a revised strategy focused on comparative evaluation using models that could be reliably tested within the available resources.

This decision reflects sound experimental judgment and aligns with best practices in scientific research.

9 XDoG: A Non-Learning Bio-Inspired Baseline

Extended Difference of Gaussians (XDoG) is a bio-inspired edge detector derived from classical DoG filtering. It simulates center-surround receptive fields and applies non-linear enhancement to emphasize perceptually meaningful contours.

XDoG requires no training, making it computationally efficient and highly interpretable.

10 DEXINED: A Pretrained Bio-Inspired Deep Model

DEXINED (Deep EXplicit INspired Edge Detector) represents a hybrid approach that integrates biological principles into a deep learning framework. Unlike XYW-Net, DEXINED provides pretrained models, enabling direct evaluation.

The architecture emphasizes explicit edge-awareness and multi-scale fusion, producing thin and well-localized edges.

11 Reference Deep Learning Models

To contextualize the results, two standard deep learning models were used:

- HED (Holistically-Nested Edge Detection)
- RCF (Richer Convolutional Features)

These models represent strong baselines in modern edge detection research.

12 Experimental Design

The final experimental setup compares:

- XDoG (bio-inspired, non-learning)
- DEXINED (bio-inspired, pretrained)
- HED and RCF (standard deep learning)

The evaluation emphasizes structural quality, robustness, and qualitative behavior rather than raw metric optimization.

13 Discussion

The results and analyses highlight the strengths and weaknesses of each paradigm. While purely bio-inspired deep models remain challenging to train, hybrid and pretrained approaches demonstrate strong potential.

Bio-inspired models introduce inductive biases that can reduce reliance on massive datasets and improve robustness.

14 Future Directions

Future work may explore:

- Federated or distributed training of bio-inspired models
- Hybrid architectures combining retinal and cortical principles
- Energy-efficient edge detection for edge devices

These directions align with both biological insight and modern AI research.

15 Conclusion

This semester-long project provides a comprehensive investigation of bio-inspired edge detection. While full training of XYW-Net was not feasible, the project successfully analyzed its limitations and redirected focus toward viable alternatives.

The comparative study of XDoG, DEXINED, HED, and RCF demonstrates that bio-inspired vision remains a promising and necessary research direction for the future of computer vision.